Linked Micromap Plots in R

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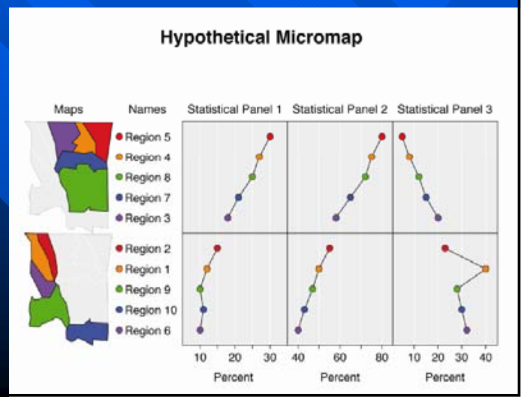
IASC Satellite Conference, Seoul, Korea -- August 22, 2013

Contents

- Background
- Stand-alone Approaches
- Templates in Support of Carr & Pickle (2010)
- R Package micromap
- R Package micromapST
- Conclusion & Outlook

Micromaps

- Link of row-labeled univariate (or multivariate) statistical summaries to corresponding geographical region
- Focus on statistical display and not on maps
- Useful for
  - environmental data
  - agricultural data
  - medical data
  - economical data

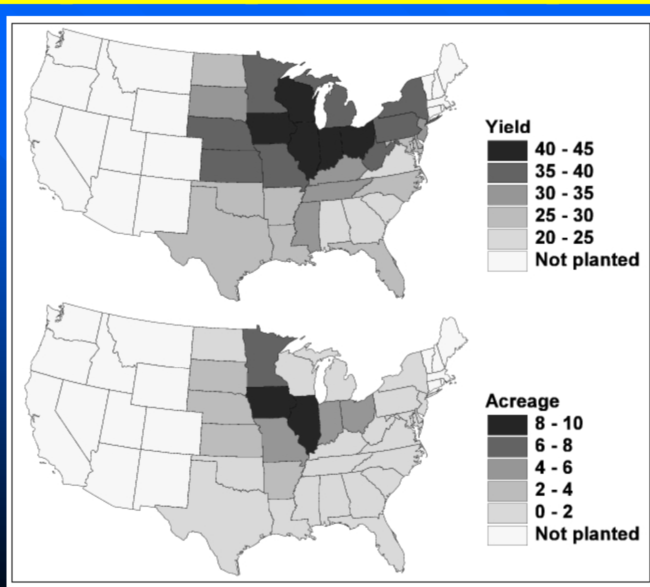


History of Micromaps

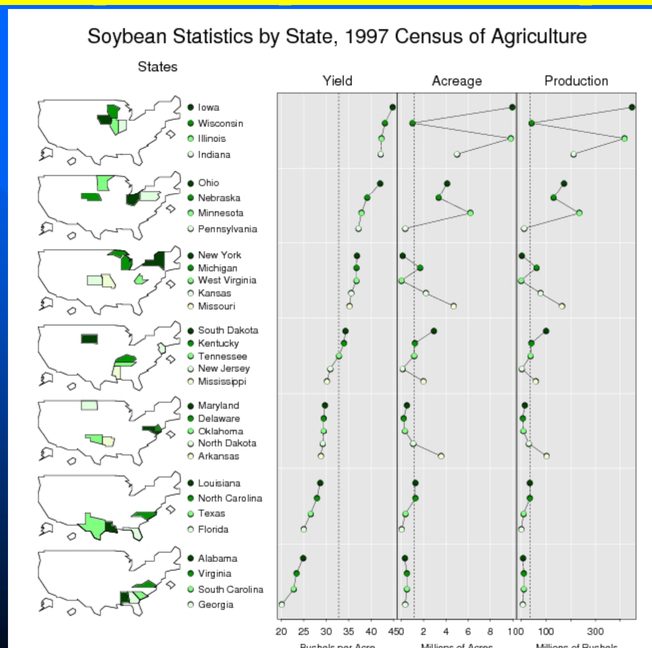
- First presented at 1995 American Statistical Association's annual meeting (Olsen, Carr, Courbois, Pierson)
- Initial references:
  - Carr, Pierson (1996) Emphasizing Statistical Summaries …with Micromaps, SCSG*, Vol.7, No.3
  - Carr, Olsen, Courbois, Pierson, Carr (1998) Linked Micromap Plots …, SCSG, Vol. 9, No.1

*Statistical Computing & Statistical Graphics Newsletter: http://stat-computing.org/newsletter/archive.html

Choropleth Maps vs Micromaps (1)



Choropleth Maps vs Micromaps (2)



Creation of Micromaps

- Primarily created in S-Plus and Java
- In 2006, Symanzik & Carr (2008) stated:

  "It seems to be possible to adapt the S–Plus micromap code for use in the R statistical package (Ihaka and Gentleman, 1996), which can be freely obtained from http://www.r-project.org/. Although no full implementation of a micromap library in R exists at this point, the basic panel functions and simple examples of LM plots have been converted from S–Plus to R by Anthony R. Olsen."

Stand-alone Approaches

- Mike Minnotte (~2006/2007):
  - Created basic micromap script for his graduate–level "Graphical Methods" course at Utah State University (USU)
  - Based on R command "layout" and R package "maps"
- Brian Diggs (2011):
  - Sample code, posted online at https://groups.google.com/forum/?fromgroups#!topic/ggplot2/djLY7AeCd7Y
  - Based on R packages "ggplot2" and "maps"
- Nathan Voge (2011):
  - Simplified, documented, and extended existing R code for his MS Project at USU

Templates in Support of Carr & Pickle (2010)

- R functions and scripts for linked micromap plots and comparative micromap plots
- Freely accessible at http://mason.gmu.edu/~dcarr/Micromaps/
- Central part of the implementation:
  - Set of panel functions, such as panelLayout, panelSelect, panelScale, panelFill, and panelGrid
  - Allows users to specify details of the overall plotting area of each component of the plot — and then draw the desired subplot into this plotting area
  - User has to loop through each of the various panels (for maps, identifiers, and statistical data) and each of the rows of the data and explicitly indicate which information (dots, names, regions in a map) has to be plotted in that panel

R Package micromap

- Developed by Payton, McManus, Weber, Olsen, and Kincaid (2012/2013)
- Based on R package "ggplot2"
- Not necessary to loop through each column and row of the data to create a cell in the LM plot
- Rather, main functionality is hidden in two main functions: "lmplot" and "lmgroupedplot"

R Package micromapST

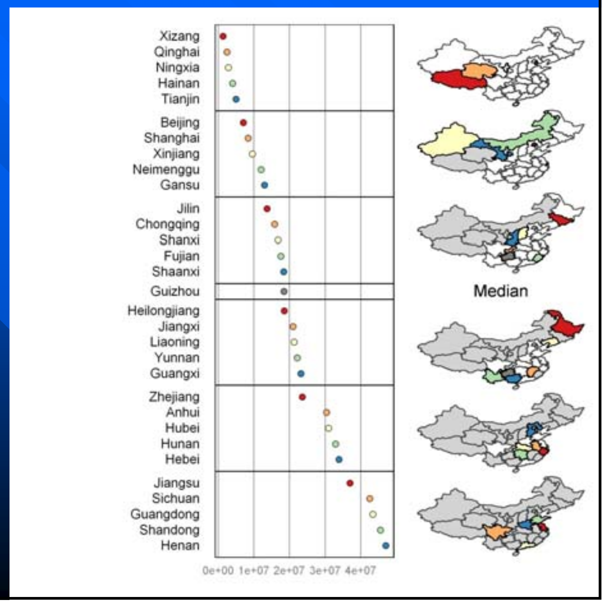
- Developed by Carr, Pearson Jr., and Pickle (2013)
- Focuses exclusively on special linked map design for U.S. states
- Input simplification replaces writing or editing a long script
- Two required input arguments are:
  - Data frame with a row for each state
  - Panel description data frame with a row for each column in the linked micromap

Concluding Remarks

- R packages micromap & micromapST make it easy to create "standard" micromaps
- For specific purposes, it may still be necessary to directly adapt the code from Carr & Pickle (2010)

Ongoing Work

- Regional Micromaps:
  - Korea: see next talk by Jeong Yong Ahn
  - China: work in progress (with micromap R package)



- **References:**

Carr, D. B., Pickle, L. W. (2010): Visualizing Data Patterns with Micromaps, Chapman & Hall/CRC, Boca Raton, FL.

Symanzik, J., Carr, D. B. (2008): Interactive Linked Micromap Plots for the Display of Geographically Referenced Statistical Data, In: Chen, C., Härdle, W., Unwin, A. (Eds.), Handbook of Data Visualization, Springer, Berlin/Heidelberg, 267-294.


Questions ???